STANDARD OPERATING PROCEDURE FOR THE OPERATION OF THE AGILENT 7890B and 8890 AUTOMATED GAS CHROMATOGRAPH SYSTEM AND ASSOCIATED COMPONENTS.

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Issue Date: _____ **Effective Date:** _____

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1.0 PURPOSE

This Standard Operating Procedure (SOP) delineates the steps required to operate, calibrate and collect the data associated with the Markes CIA Advantage-xr, Kori-xr, UNITY-xr, and Agilent 7890b and 8890 automated gas chromatograph (AutoGC) systems and their associated components. By following this SOP, the operator ensures that through proper operations, calibrations, and maintenance, the AutoGC systems produce the appropriate quantity and quality of data required to meet data quality objectives (DQOs) set forth in the *Photochemical Assessment Monitoring Station (PAMS) Network Quality Assurance Project Plan (QAPP) for Utah Air Monitoring.*

2.0 SCOPE AND APPLICABILITY

2.1 Intent for Use

This procedure is intended for use by the Air Monitoring Section's AutoGC site operators. It is assumed that an AutoGC site operator has a technical background with knowledge of air pollution, federal and state environmental law, and monitoring programs within the DAQ. It is also assumed that an AutoGC site operator has general knowledge (via training) of operations of the Markes Instrument Control and Agilent EZChrom software environments, ORSAT Merlin AutoGC Xplorer (MAX), cloud-based data management software, and has the required user accounts, permissions, and tools needed to perform the functions outlined in this SOP.

2.2 Valid Time frame

The SOP covers the timeframe between the effective date on the original copy until superseded by another approved SOP. Previous versions of the original hardcopy SOP are archived for future reference.

3.0 METHOD SUMMARY

On an hourly cycle, the Markes CIA Advantage-xr, Kori-xr, UNITY-xr, and Agilent 7890b (system using Helium as a carrying gas) or Agilent 8890 (system using Hydrogen as a carrying gas) AutoGC system draws in ambient air from the inlet probe to collect and preconcentrate VOCs from the sampled atmosphere and subsequently separate the VOCs for detection via a flame ionization detector (FID). A new sample commences at the top of each hour. Ambient atmosphere is drawn into the Markes CIA Advantage-xr sampling inlet by the vacuum supplied by the system's sampling pump and routed through a blank quartz trap within the Kori-xr maintained at -30ºC by Peltier cooling. Sample collection occurs for 40 minutes and flow is controlled at 20 mL/minute by a mass flow controller to acquire 800 mL of sample. The sampled atmosphere within the Kori-xr trap is

cooled to remove moisture prior to passing the sampled atmosphere to the UNITY-xr trap. The two-phase sorbent trap within the UNITY-xr is desorbed by quickly heating (40ºC/minute) to 325ºC which extracts and traps the VOCs while purging excess air. Following desorption, the trap is then backflushed with carrier gas to sweep the desorbed VOCs from the trap for separation in the GC.

Within the GC, the compounds pass through the Boiling Point (BP) separation column and are routed through a Dean's switch, which directs the lighter hydrocarbons (C2-C6) through a subsequent alumina Porous Layer Open Tubular (PLOT) separation column. The PLOT column separates the lighter hydrocarbons by weight as they pass through before being ignited by a hydrogen-fueled flame ionization detector (FID). Compounds generally elute from lighter to heavier by weight, but many factors such as flow, humidity, and temperature can affect the elution of PLOT column compounds. The FID detects ions produced by the ignition of compounds and translates them to an electronic signal. Following the elution of 1-hexene, the Dean's switch flips and heavier compounds (C6-C12) pass solely through the BP column and then a restrictor for separation. These heavier compounds elute to a separate FID from lower to higher boiling points which is directly correlated to lighter to heavier weights respectively.

Compounds are identified by their retention time in the analytical method within the Agilent EZChrom software. The amperage responses of the FIDs are converted to a concentration in parts per billion carbon (ppbC) based on the response factor according to the calibration. The response factor is based on the FID response of propane or butane (C2-C6) or benzene or toluene (C6-C12), depending which is the calibrant selected for the BP column, in a multipoint calibration. Data is then stored on the site computer, transmitted to the ORSAT MAX website, and archived manually to the U:\ network drive at

U:\PLAN\AMC\GC-Agilent+Markes\Data

System functions are controlled by a connected personal computer running the EZChrom software. The AutoGC requires additional supplies and support equipment including a zero air/TOC generator with support compressor, source of helium or hydrogen carrier gas, source of hydrogen (hydrogen generator or cylinder gas) for FID fuel, and a system to dilute stock standard gasses for calibration and ongoing QC checks.

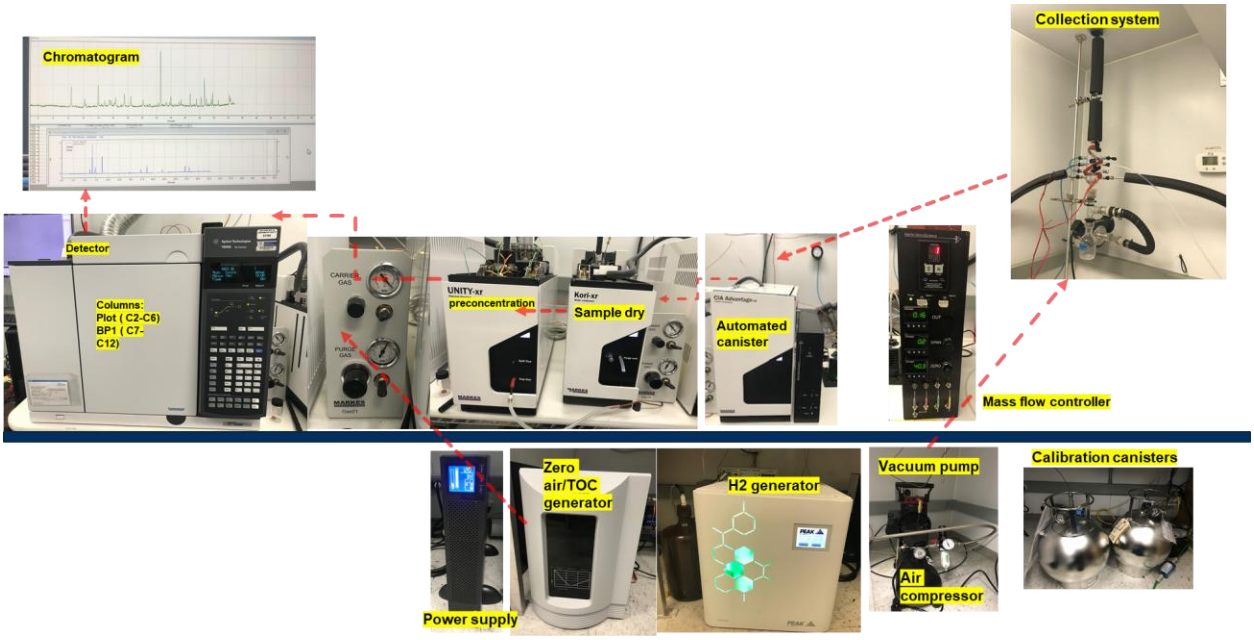


Figure 3.1 AutoGC configuration

4.0 LIMITATIONS

Certain limitations exist that may interfere with AutoGC operations and data collection.

The AutoGC system subject to this SOP operates at a remote sampling site that samples ambient air. The quality of the data produced by the AutoGC system can be affected by changes both inside and outside of the site building. The site may be subjected to power failures, air conditioning/heating failures, and data connection failures. Changes in ambient weather conditions, especially large fluctuations and temperature and humidity, can affect the data. Other factors that can affect the AutoGC include water being introduced into the sample flow path and other hardware failures.

Operators must be aware of the large amount of data produced by the AutoGC system. Frequent contact with the instrument to ensure proper operation, including monitoring for QC sample failures, system failures due to malfunctions or power outages, etc. can jeopardize the validity of large amounts of data which can quickly lower the percent data recovery for the site. Operators are encouraged to check in on the system at least daily remotely and weekly on site, make accurate electronic logs of station activity, monitor QC samples, ensure future sample collections are scheduled and queued properly in the sequence, and follow all preventive maintenance schedules.

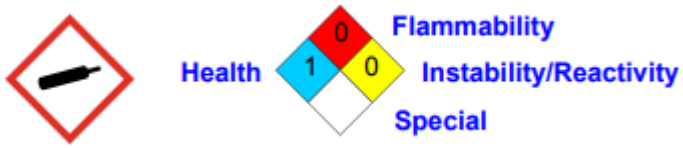
5.0 SAFETY

Operation of an AutoGC system presents several safety hazards of which site operators and site visitors must be aware.

The FIDs in the GC utilize hydrogen gas as fuel. Leaks in the hydrogen delivery system may result in hydrogen build up which can cause a fire or explosion. Instruments should not be operated near an open flame, and sources of sparks should be eliminated. The shelter in which the instrument is installed should have a hydrogen gas detection/warning system.

Hydrogen Cylinder

Standard gasses and support gasses are contained within high pressure cylinders or canisters and can include volatile organic compounds (VOCs). Staff working with high pressure cylinders must be properly trained on the proper handling and operating aspects.



Helium and Standard Cylinders

Operators should exercise caution when working on interior components of the instruments as internal components operate at both high temperatures (GC oven, FID, trap during desorption) and very cold temperatures (Peltier cooled trap) which can cause burns to unprotected skin. Operators should allow internal parts to return to near-ambient temperatures before handling. Failure to do so can result in severe burns.

Silica gel is used as a desiccant in the Hydrogen generator. The SDS sheets for all safety hazards are located on the GC desktop computer in a folder labeled SDS Sheets.

6.0 EQUIPMENT, SUPPLIES, REAGENTS, AND STANDARDS

The system is composed of the Markes components, the Agilent 7890 B or 8890 GC with dual FID, and additional support equipment. The Markes system includes the UNITY-Xr TD, Kori-Xr water abstraction device and the CIA Advantage-xr gas sampling inlet.

6.1 Equipment

Agilent 7890B GC

- Microfluidic Deans switch
- Front and Rear flame ionization detectors (FIDs)
- Agilent 50 m PLOT Al2O3-Na2SO4 0.32 mm ID column with an 8 µm film thickness or equivalent
- Agilent 50 m PDMS-coated durabond-1 0.32 mm ID capillary column with a 1.05 µm film thickness or equivalent
- Transfer line

Agilent 8890 GC

- Microfluidic Deans switch
- Front and Rear flame ionization detectors (FIDs)
- Agilent 50 m PLOT Al2O3 0.32 mm ID column with an 8 µm film thickness or equivalent
- SGE 50m BP1 Dimethyl Polysiloxane 0.22mm ID columns with a film thickness of 1µm
- Transfer line

Markes UNITY-xr TD system

- Markes CIA Advantage-XR with 16-port heated sampling interface for the Helium carrier systems and with 4-port heated sampling interface for the Hydrogen carrier gas systems
    - Heated transfer line to Kori-XR
- Markes Kori-XR water management device
    - Empty cold glass trap (P/N MKI-U-T1KORI or equivalent)
    - Heated transfer line to Unity-XR
- Markes Unity-XR TD system- Preconcentrator trap
- PAMS (VOC O3 precursors) trap (P/N MKI-U-T20PAM or P/N MKI-U-T17)
    - Heated transfer line to GC
    - Sampling Pump – KNF model N86KN.18 or equivalent

Additional Support equipment

- Parker Balston TOC gas generator (TOC-1250NA)
- PEAK Scientific hydrogen generator for 7890b (63-0100)
- Parker hydrogen generator for 8890 (H2PEMPD-510-100)
- Merlin MicroScience gas dilution system

- A helium leak detector (GL Sciences LD239)
- A mass flow meter
- A Windows-based personal computer that provides:
  - An ethernet or internet connection capable of providing access to DAQ's internal file system
  - Markes Instrument Control software
  - Agilent EZChrom software
  - Merlin EZSequence software
  - An internet browser
  - Access to electronic operator logbook and checklist on Google Docs/Sheets
  - An ORSAT MAX account
  - A File Transfer Protocol (FTP) software (such as FileZilla)
  - A program that automatically uploads data files to ORSAT MAX
  - 7-ZIP, WinZip, or equivalent file extraction and compression software
  - Text editor software (such as WordPad or TextPad Editor)
  - Microsoft Excel
  - Adobe Acrobat Reader or other PDF file reader
  - TeamViewer

6.2 Supplies

Consumable supplies that are required to be in stock and replaced regularly are listed below by equipment application.

Consumable Supplies for the PEAK Scientific Hydrogen Generator

- Silica gel beads (can be reused through oven bake out)
- Deionized water

6.3 Reagents and Chemicals

- Calibration Verification Standard (CVS) –14-component (see Table 1), 1 ppmV NIST-certified or NIST-traceable certified primary standard gas.
- Laboratory Calibration Standard (LCS) – 14-component (see Table 1), 5 ppbV diluted from 1 ppmV NIST-certified or NIST-traceable certified second source standard gas.
- Retention Time Standard (RTS) – 57-component, 8 to 60 ppbC gas mixture.
- Blank – Zero air or hydrocarbon-free air containing nominally ≤ 0.05 ppbC total hydrocarbons provided by a zero-air generator or cylinder.

- Hydrogen (H_2) gas: Hydrogen generator or UHP grade cylinder gas
- For 7890B GC: UHP grade cylinder Helium (He) gas

Table 1. List of calibrant compounds present in the CVS and LCS.

PLOT Column *(Listed in elution order)*	**Boiling Point (BP) Column** *(Listed in elution order)*
Ethane	n-Hexane
Propane	Benzene
n-Butane	Toluene
Acetylene	m&p-Xylene
n-Pentane	n-Propylbenzene
1,3-Butadiene	1,2,4-Trimethylbenzene
2-Methylpentane	p-Diethylbenzene

7.0 PROCEDURE

7.1 Background

This section describes in detail the procedures for operating and maintaining DAQ's hourly AutoGC analyzers.

AutoGCs are scheduled to operate 24 hours a day throughout the year with the majority of hours reserved for ambient sampling. Other hours are reserved for quality control runs or are interrupted by instrument maintenance. The instruments are designed to identify target volatile organic compounds (VOCs) that are of interest primarily because of their contribution to ground-level ozone formation and/or toxicity. Appendix A, the AutoGC Master Compound List, includes current as well as historical target VOCs as required by the DAQ PAMS QAPP.

The AutoGC operator verifies that the target compounds are identified, quantified, and reported according to the DAQ PAMS QAPP. Operators and validators must communicate with each other to maximize data quality and data capture.

7.2 Instrument Setup and Installation

AutoGC

Agilent or a contractor will typically perform the initial setup and site installation of the AutoGC system. The AutoGC is to be installed in a shelter maintained between 15-30°C and should be acceptable for other types of monitors installed in the shelter as well. The required power supply for the Markes UNITY-xr TD and Agilent 7890B/8890 GC is a 20 A 110 V AC circuit. Additional circuits are necessary to supply power to the zero-air generator, compressor, computer, hydrogen generator, CIA Advantage-xr, and Kori-xr. The GC should not be installed on a UPS as a GC requires a large output sine-wave UPS and current available UPS devices that match requirements for a GC would only last approximately 20 minutes before the power is cut resulting in invalid data. The computer is recommended to be on a UPS.

- Open all gas cylinders and set the zero air and helium to maintain an outlet pressure of 80-90 psi (for the 7890B system).
- The hydrogen cylinder outlet pressure on the manifold should be 70 psi. The hydrogen generator should be set to 90. Once the H2 generator valve is open, the outlet manifold pressure will increase from 70 to 90 psi.
- Open Markes system regulators and set the carrier gas pressure to 60 psi and the purge gas pressure to 60 psi (7890B system).
- The 8890 **Gas 1** carrier should be 50 psi, purge 50 psi. **Gas 2** carrier 20psi, purge 60 psi.
- Power on all systems: computer, monitor, GC, Markes front-end and pump.

  Note: The GC power button is located on the lower left front corner of the GC. The Markes CIA-Advantage, Kori and Unity power switches are in the back of the units. The Markes system pump power switch is the green switch located on the front of the pump.

- The GC will try to auto-ignite the FID detectors after they warm up. This can take various minutes to happen. A crackle sound will be heard when the FID is lit. To check the FID ignition manually on the 7890b, use the GC keypad panel: select either “Front Det” for the front detector or “Back Det” for the rear detector. Use the up and down arrows to navigate to H2 flow, air flow and make up flow. Press on/yes and then enter to check the flow if it is not already running. Use the up and down arrow to navigate to flame and press on/yes and enter. The system will switch from off to igniting. Once a popping noise is heard the flame has ignited and the detector will switch to on mode. Repeat the steps for the detector not selected previously. Once both FIDs are ignited the system can be used for analysis after a 30-minute equilibration period. The 8890 has a touchscreen display with the menu on the left that will direct to the detectors, columns, oven, etc.
- Connect the AutoGC inlet line to port #1 (unless otherwise instructed) of the CIA-Advantage heated interface. Document this connection in the electronic logbook.

- Connect 6-L summa canister(s) to desired port(s) with 1/16 in (0.02 in ID) tubing and appropriate fittings. Log connections to CIA-Advantage ports in the site electronic logbook. A summary of settings for both systems is in Table 2

Table 2. Summary of GC settings

AGILENT 7890 B	AGILENT 8890
Carrier Gas: Helium	Carrier Gas: Hydrogen
GC Run time: 48 min	GC Run time: 48 min
Front inlet · Temp: 150 C · Pressure: 38.168 psi · Flow: 5.487 ml/min · Septum purge flow: 3.0 ml/min	**Front inlet** · Temp: 150 C · Pressure: 28.085 psi · Flow: 3.5 ml/min · Septum purge flow: 1.0 ml/min
Front Detector/ Back Detector · Temp: 250 C · Fuel flow: 35 ml/min · Air flow: 350 ml/min · Make up flow: 10 ml/min · Signal value: 2.1 PA	**Front Detector/ Back Detector** · Temp: 250 C · Fuel flow: 30 ml/min · Air flow: 300 ml/min · Make up flow: 0 ml/min · Signal value: 2.9 PA
Column flows Column 1: 4.5 ml/min Column 2: 3 ml/min	**Column flows** Column 1: 4 ml/min Column 2: 1.5 ml/min

Markes Instrument Controller (MIC)

- Open Markes Instrument Controller: MIC 2.0
- A dialogue box will appear and the software will communicate with the CIA-Advantage, Kori and Unity and when the communication ends the MIC 2.0 home screen will be displayed (see Figure 1)

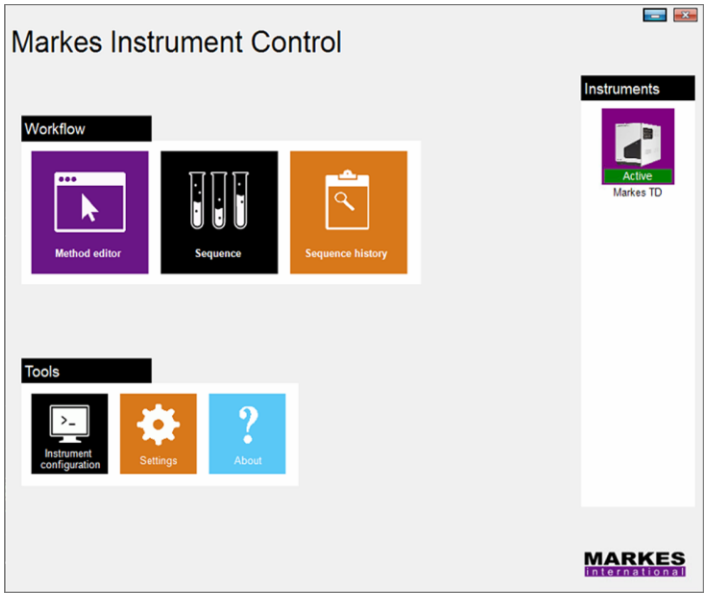


Figure 1. MIC 2.0 Home Screen

- The method editor option allows the creation or modification of the method used for trapping the VOCs. The default method is named PAMS_AutoGC.
- The sequence option allows you to create a sequence. We will only use the "TD - Orsat PAMS H2" method. After hitting play it will bring up a new window. click the top box (run a sequence of 1 time, continuous). (Figure 2) provides the information about the status of the system

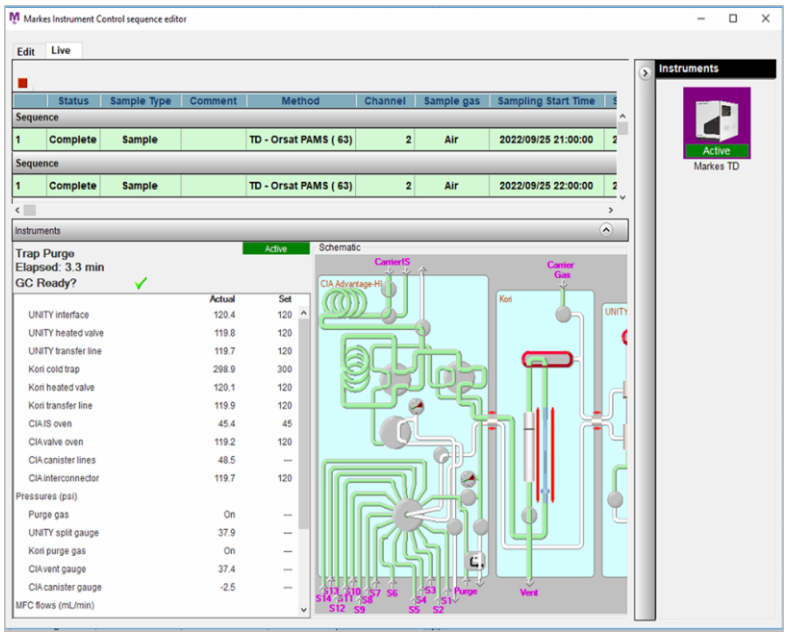


Figure 2. MIC 2.0 Sequence Table

- The sequence History option allows you to check past sample information (Figure 3)

Sequence

Run #	Status	Sample ID	Method	Filename	Action	Description
124	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23D.dat		10/23/2022 03:00
125	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23E.dat		10/23/2022 04:00
126	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23F.dat		10/23/2022 05:00
127	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23G.dat		10/23/2022 06:00
128	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23H.dat		10/23/2022 07:00
129	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23I.dat		10/23/2022 08:00
130	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23J.dat		10/23/2022 09:00
131	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23K.dat		10/23/2022 10:00
132	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23L.dat		10/23/2022 11:00
133	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23M.dat		10/23/2022 12:00
134	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23N.dat		10/23/2022 13:00
135	Complete	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23O.dat		10/23/2022 14:00
136	Acquiring	Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23P.dat		10/23/2022 15:00
137		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23Q.dat		10/23/2022 16:00
138		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23R.dat		10/23/2022 17:00
139		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23S.dat		10/23/2022 18:00
140		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23T.dat		10/23/2022 19:00
141		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23U.dat		10/23/2022 20:00
142		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23V.dat		10/23/2022 21:00
143		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23W.dat		10/23/2022 22:00
144		Ambient Air	V-meth\Orsat_PAMS.met	HWSJ23X.dat		10/23/2022 23:00

Figure 3. MIC 2.0 Sequence Table

Methods

There are two types of methods: operational methods that control the hardware operations of the AutoGC system and analytical methods that control how the chromatography data is analyzed. On the Markes side, there is only an operational method, whereas on the Agilent side, there is a separate combined operational and analytical method. AutoGC operators should be able to modify all methods, while data validators can only modify the analytical side of the Agilent method in EZChrom.

The Markes operational method should generally not be modified unless instructed or recommended to do so. It controls parameters such as flow rates, temperatures, and timing within the CIA Advantage-xr, Kori-xr, and UNITY-xr. The typical method used is the "TD-Orsat PAMS" method. If troubleshooting, other built in methods may be more appropriate especially if a shorter sample or cycle time is needed. Every time the Unity trap is changed it will need to be conditioned. The method "T17/T20 Trap Conditioning" will be selected. To modify any method, navigate to the "Method Editor" in Markes Instrument Control. Usually any modifications done to the Markes method will be to compensate for changes in weather that significantly affect compound analysis, following maintenance on the Markes instruments, or to provide for better extraction of target compounds following further research.

The Agilent GC method, controlled through EZChrom, requires much more operator modification particularly on the analytical side of the method. The two most common operational method reviews or adjustments to the Agilent GC method are:

- Dean's Switch Cut Time – this is the time after the start of the GC run when the Dean's Switch changes from flowing through both the BP and PLOT columns to

FID A to only flowing through the BP column to a restrictor that goes to FID B. The cut time should fall between the elution of 1-hexene and n-hexane. If one of either 1-hexene or n-hexane partially or fully elutes on the incorrect column, the cut time should be adjusted. Change it by going to Method -> Instrument Setup -> Agilent 7890B/8890 -> Events -> Line 6 for Valve1

- Column Information – when a column is changed inside the GC, the information on the column and within the method need to be compared to ensure accuracy. To review column information in EZChrom, go to Method -> Instrument Setup -> Agilent 7890B/8890 -> Columns …and Method -> Instrument Setup -> Agilent 7890B/8890 -> Configuration -> Columns

  There are other settings such as for the oven heating ramp, detectors, flows, signals, events, and readiness that may have to be edited during troubleshooting or if the software needs to be reinstalled. Consult with ORSAT, Agilent, or others prior to adjusting these settings.

  The Agilent GC analytical method is where the vast majority of data analysis occurs. The EZChrom software processes the chromatography data, calculates pico amperage values, integrates peaks, calculates peak areas, converts peak areas to concentration values, and assigns compound names to peaks.

7.3 Routine Operations

Daily Tasks

- Log in to ORSAT MAX website and check on file upload status. Missing files (red cell Figure 4) or the upload status not being current could indicate a problem with communications or the AutoGC. The upload software pulls the data to MAX every three hours so it is common to not have the most current files in MAX yet.

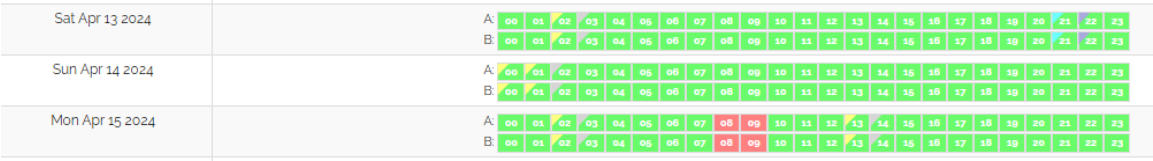


Figure 4. ORSAT MAX File upload status main screen

Data Verification

On a daily basis, operational staff will use ORSAT MAX software to review the QC checks and briefly inspect the daily trends to identify possible issues like missing data or mismatched hours. A general comment about the daily QC checks is recorded on a google sheet, here is an example of the daily QC tracking sheet:

https://docs.google.com/spreadsheets/d/1BGAQRJoxqxaGMqerwyUtnSbGitDMZoCCtDqIp_tVogc/edit?usp=drive_link

ORSAT MAX, check daily CVS recoveries and Blank results:

- Log in to ORSAT MAX website
- Select the current day or the date that you want to check the recoveries
- Scroll down and look for the CVS recovery table for each column and verify that the recoveries are within the expected range (70-130%). Document the compounds and the percent recovery if they are out of range.
- If the daily CVS didn't run or if the recoveries for Propane / Butane in the PLOT column or Benzene/ Toluene in the BP column are out of the desired range schedule another CVS run

On a daily basis, operational staff will use ORSAT MAX software to review the Daily Zero checks to inspect if any compounds came in above the MDL. The zero's for all compounds are pulled from MAX via csv file. It is located in the Reports icon in the top right corner. There is an option for crosstab CSV. Once in you can then input the site, date, report output option, and QC option. It is then transposed into a xlsx and pasted into the HW_BLANK Checks file located on the drive.

https://docs.google.com/spreadsheets/d/16TBjvRY76QUMoI72qvrVFSV44_v0lWzH4fVAILV0E5c/edit?gid=0#gid=0

On a weekly/monthly basis, Technical staff will use will use ORSAT MAX to verify the proper instrument operation, the review consist of checking the time Series plots (Ambient Air), Scatter Relationships and Diurnal Chats for the most abundant VOC species collected as part of the PAMS program**. Figure 9** displays an example of the plots used to review the ambient data using ORSAT MAX.

- On The GC computer,

- Log in to the site computer and ensure the Markes and GC sequences are running properly
- Open the offline version of EZChrom, open a representative ambient hour, and check the chromatography and method
- Communicate with data validators if problems arise
- Document changes in site operator log

Figure 9. example of the plots used to review the ambient data using ORSAT MAX.

Some examples of the regular checks include:

- Inspect that all the abundant species (ethane, propane, n-butane, iso-butane, iso-pentane, n-pentane, n-hexane, benzene, toluene, xylenes and ethylbenzene) are present in all the hours.
- check that: propane > propylene; ethane> ethylene; n-butane >Iso-butane; 2-methylpentane > 3-methylpentane > 2,3-dimethylbutane; 2-methylhexane> 2,3-dimethylpentane; methylcyclopentane > 2,4-dimethylpentane; 1,2,4-trimethylbenzene> than 1,2,3-trimethylbenzene and 1,3,5-trimethylbenzene; m-ethyltoluene> p-ethyltoluene and o-ethyltoluene.
- Review that all the pentanes trend and the iso-pentane should be the highest
- Review that all the hexanes trend and n-hexane should be the highest
- Review that all the heptanes trend and the n-heptane should be the highest
- Review that the normal alkanes C2-C5 trend as: ethane/propane> butane> pentane
- Review that the normal alkanes C6-C11 trend as: hexane> heptane> octane> nonane> decane> undecane
- TNMTC< TNMHC normally about 15%

Note that if some of these species are missing or some expected patterns are not followed, it is not necessarily an indication of instrument failure. It can be the result of misidentification or a partially direct source(s) affecting the site. It will require performing additional checks of the common patterns and fingerprint plots to drive a conclusion.

Weekly Tasks

- Perform a site visit
  - Complete site operator checklist (see Table 3)
  - Fill Hydrogen Generator reservoir with DI water

Table 3. Operator checklist

Date	1/3/2024	1/9/2024	1/15/2024	1/25/2024	1/29/2024	2/7/2024	2/15/2024	2/20/2024
Time	15:38	13:09	13:46	13:18	11:45	15:25	14:55	11:42
Operator	LB	LB	LB	LB	LB	LB	LB	LB
H2 Generator Alarm	no	no	no	no	no	no	no	no
H2 Generator, pressure, psi	103	103	103	103	103	103	103	103
H2 Generator Set Point	103	103	103	103	103	103	103	103
H2 Generator, reservoir (3.78L=full)	half. Filled	3/4. filled	3/4. filled	2/3. filled	full	half. Filled	2/3. filled	3/4, filled
Manifold Blower working?	yes	yes	yes	yes	yes	yes	yes	yes
Air compressor, ballast pressure, psi	80	92.0	88	85	94	84	100	81
Air compressor, outlet pressure, psi	83	85	85	85	85	85	85	82
Compressor/TOC Regulator Drip Status	dry	dry	a touch damp	a touch damp	dry	dry	dry	dry
Empty water from air compressor moisture catch?	yes	yes	yes	yes	yes	yes	yes	yes
TOC mass flow meter	0.31	0.33	0.33	0.33	0.32	0.31	0.32	0.33
Inline Reg	69	75	74	73	74	73	73	72
Zero Air Generator pressure, psi	72	79	77	76	77	75	74	74
Kori Purge	87ccm ->95ccm							
Gas-01 Carrier	55	56	56	56	55	56	56	56
Gas-01 Purge	54	53	53	53	54	54	54	54
Gas-02 Carrier	20	20	20	20	20	20	20	20
Gas-02 Purge	49	48	48	48	48	48	48	48
CVS								
15L CVS CC524271-0325 can 2451	19.5	18	17.3	16	15.5	14.5	13	12, removed
15L CVS CC524271-0325 can 3695								30.5
15L LCS CC177206-0125 can 3733					installed, 29	28.4	27	26.2
15L LCS CC177206-0125 can 3187	6.9	5.6	4.7	3.9	3, removed	-		
RTS cylinder/canister, psi	11	11	11	10.5	10.5	11	11	10
RTS CYLINDER VALVE CLOSED								
TOC Catalyst	warm	warm	warm	warm	warm	warm	warm	warm
Zero Air Catalyst	warm	warm	warm	warm	warm	warm	warm	warm
Magnahelix-	0.1	0.1	0.5	0.5	0.2	0.5	0.5	0.5
Sample lines	warm	warm	warm	warm	warm	warm	warm	warm
Kori Purge Flow								
Blender Span	5.2	5.1	5.1	5.1	5.2	5.1	5.2	5.1
Blender Zero	39.9	39.8	39.8	39.7	39.8	39.7	39.7	39.7
Is blender Span valve shutoff?	yes	yes	yes	yes	yes	yes	yes	yes
Blender Output	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12
Station Temp-	18.6	20.5	19.6	19	19.5	20.7	19.8	19.3
Current Sequence Row	297	445	110	349	444	184	375	492
Unity Sample Start Time	15:00	12:00	13:00	13:00	11:00	15:00	15:00	12:00
Ambient Weather	cloudy cold inverted	cold,cloudy, storm incoming	sunny, breezy, nice	cloudy, rainy, cool	sunny, a bit inverted, warm for Jan	snowy/rainy/calm	sunny, showery	cloudy, showery
Outside Relative Humidity	72.8	41.8%	48%	94%	62%	98%	63%	80%
Outside Temperature	2.2	-0.9	-2.1	2.2	7.5	0.5	5.4	5.9
Max/min Rtemp							21.6/16.8	21.4/17.1
Notes		adjusted AC setpt from 74 to 75						

- Review and complete site operator log
- Clean monitoring shelter (do not use any liquid cleaners other than deionized water)
- Troubleshoot problems that arise

- On ORSAT MAX, check the weekly LCS recoveries, RTS peak identification as well as the precision checks (CVS1 and CVS2). To review the Precision (CV1 and CV2) and LCS, follow the same procedure performed to check the daily CVS recoveries
- **Note:** For the precision, verify that the RPD % value is <= 25. Usually Acetylene and P-Diethylbenzene have lower recoveries. If the weekly precision check and /or LCS didn't run or if the recoveries for Propane/ Butane in the PLOT column or/and Toluene in the BP column are out of the desired range, schedule another precision/ LCS run(s). The RTS check passes if all the compounds, with few exceptions (e.g. acetylene), are identified in the run. Sometimes missing compounds in the RTS is due to misidentification of the peaks or species that have been removed from the standard but are still in the list of target compounds (e.g. pinenes)

- Review analytical method on EZChrom. Update method and reprocess data files as necessary (see section 7.7 for reprocessing)

Semi-Annual Tasks

- Clean or change Kori-xr trap
- Change UNITY-xr trap
- Perform multipoint calibration

Annual Tasks

- Calculate the Manifold Residence Time for the GC (see section 7.9)

Before and After the PAMS season

- Schedule the runs to estimate the Method Detection Limit (MDL)

7.4 Data acquisition

Sequences

Sequences are spreadsheet schedules used to control the timing of the Markes and Agilent sides of the system. In the Markes-Agilent setup, two sequences are required to be created and maintained: one for the Markes thermal desorption and one for the Agilent GC.

To start a Markes sequence:

- Open the Markes MIC 2.0 software.
- In Markes Instrument Control, under Workflow, click on "Sequence".

  Note: Only when Markes is installed for the first time:

  - In the first line of the sequence spreadsheet, put "Sample" for Sample Type, choose the appropriate method for Method (Usually this will be "TD – Orsat PAMS" for typical instrument operation, but a different

method may be needed while troubleshooting), choose the appropriate sample port that the inlet is plugged into in the CIA Advantage-xr for Channel, and choose "Air" for Sample gas.

- Press the Play button.
- For a typical operation, check the first box that says "Run the sequence 1 time starting at…" and make sure "Continuous" is checked. With this option, the sequence will begin preparatory work and leak checks for a sample run, then wait for the sample start time to begin at the top of the next hour. If the sample does not start, troubleshooting will be required.

To stop a Markes sequence, click the "Stop" button in the Sequence window and select the appropriate option for the timing of the stop.

The sequence for the GC is controlled through EZChrom. The EZseq.exe program on the GC computer desktop can be used to quickly create a new standardized sequence.

To create a new GC sequence:

- Click on the EZseq.exe shortcut on the desktop.
- For "Sequence Type" select "Agilent".
- Click "Apply" then click "OK" when the popup confirming the creation of the sequence file and the new sequence end date has occurred. EZseq.exe creates the sequence as a csv file. That window can then be closed.
- Open the offline site version of EZChrom (Usually a desktop shortcut with "SITE-*SiteID* (offline)" as a label.
- Go to File -> Open -> Sequence. Navigate to the D:\EZ*SiteID*-other\ folder. The new sequence will be in a filename format of *YYMMDD-SiteID*.csv
- Once open, go to File -> Save As -> Sequence. Save the new sequence in the D:\EZ*SiteID-meth* folder with the format *YYMMDD-SiteID*.seq . EZChrom will use the *SiteID*-meth folder file as the primary with the *SiteID*-other used as a backup in case the primary sequence file is corrupted.

To stop a GC sequence:

- In the online version of EZChrom, Navigate to Control -> Stop Run.
- Choose the appropriate radio button for the situation. If you are troubleshooting, usually "Stop current run and sequence run." should be chosen. "Stop current run" can be used if the system sample time is behind an hour and needs to be reset. If loading a new or modified sequence, click "Stop sequence after the current run completes." to avoid losing an hour of data for a sequence reload. This will require waiting for the run to end and for EZChrom to enter Result Review Mode.

To load the new GC sequence:

Note: If a sequence is currently running, it will need to be stopped prior to submitting the new sequence

- Go to the SITE-*SiteID* (online) version of EZChrom.
- Navigate to Control -> Sequence Run.
- For "Sequence Name" use D:\EZ*SiteID*-meth*YYYYMMDD-SiteID*.seq.
- For "Result Path" use D:\Current.
- For "Result Name" use *YYYYMMDD-SiteID-BeginLineNumber*

  where the date corresponds to the sequence date and the line number corresponds to the line number for the sample hour which the sequence is starting at
- Under "Run Range" select "Range" and enter *BeginLineNumber*-576 where the beginning line number corresponds to the sample hour that should be analyzed first

Note: It's important that the BeginLineNumber consists of 3 digits, typically when you start a new sequence your starting line number will be <100, for example starting line 11 needs to be 011

- For "Bracketing" choose "None". All other selections in the window should be blank or grayed out
- Click "Submit". The GC will now wait for the Markes to trigger the beginning of the run following sampling and thermal desorption

To start a new or modified GC sequence when EZChrom is in Result Review Mode:

- Navigate to File -> Open -> Method. Open the appropriate method file (usually Orsat_PAMS.met). This gets EZChrom out of Result Review Mode
- Navigate to File -> Open -> Sequence and open the new or modified sequence file. Verify that the correct/modified sequence is visible in the sequence window
- Navigate to Control -> Sequence Run and follow the previous steps to submit the sequence

Multipoint Calibrations

Multipoint calibrations are required to be performed following maintenance or any system adjustment. It is recommended to perform the calibrations in spring prior to ozone season and in fall after ozone season. It is also recommended to perform the multipoint calibration overnight when ambient pollution levels are lower.

Multipoint calibrations on the AutoGC system are a scheduled sequence of four hours with different standard concentration levels. The sequence is preceded by one blank hour and followed by two blank hours.

To schedule a multipoint calibration:

- Navigate to the EZChrom offline version on the site computer
- Go to File -> Open -> Sequence and open the current sequence file to be modified
- Double-click on "Action" column cell two hours prior to the initial Blank hour
- On row 1, select the following:
    - Test: Any Condition
    - Result: Pass
    - Action: Run User Program
    - Parameter: Double-click and navigate to C:\mvalve and select Runbl.exe.
- Press "OK"
- Adjust the Sample ID column to "Blank" and adjust the file name to reflect that the sample is a Blank (B)
- Repeat the above process for the next six hours. The programs that should be scheduled in the following order:
    - Hour 2: RunL1.exe (Multipoint Calibration Level 1)
    - Hour 3: RunL2.exe (Multipoint Calibration Level 2)
    - Hour 4: RunL3.exe (Multipoint Calibration Level 3)
    - Hour 5: RunL4.exe (Multipoint Calibration Level 4)
    - Hour 6: Runbl.exe
    - Hour 7: Runbl.exe

Save the modified sequence file and close the EZChrom offline version.

- Navigate to the EZChrom online version. If a sequence is currently running, go to Control -> Stop Run and select "Stop sequence after current run completes"
- Navigate to Control -> Sequence Run. Verify the correct sequence is selected. Change the last three numbers of "Result name" and the first three of "Range" to reflect the row in the sequence file for the hour currently being sampled by the Markes system. Click "Start"

Once a multipoint calibration is complete, a review of the results is required. Open the data files in the EZChrom offline version for each multipoint calibration level. Verify that the Propane or butane and toluene peaks have been properly integrated. Document the peak areas for propane and toluene.

Navigate to the Desktop and open the excel file "MMSD-MPV-18006.xlsx". Navigate to the "Curve_Response Factors" tab. Verify that the Standard Concentrations for propane

and toluene are accurate based on the standard used for the calibration. Input the peak areas for propane and benzene into the designated columns. The spreadsheet will calculate an average response factor and %RSD based upon the results. If the %RSD for either the calibrant for the PLOT column (propane or butane) or the calibrant for the BP column (benzene or toluene) exceeds 10%, the calibration has failed and further troubleshooting and/or instrument adjustment may be required. If the %RSD is below 10%, the calibration passes and the average response factor can be input into the method.

To adjust the response factors in the method, remain in the EZChrom offline version. Navigate to the Navigation sidebar and double-click on "Peaks/Groups". Ensure that "Front Signal" is selected at the top. Under the "Named Peaks" tab, input the new PLOT column response factor under "Manual RF" and fill down. Go to the "Groups" tab and input the new PLOT column response factor for the "Plot UnID" row under "Manual RF". Repeat the process for the "Back Signal"/Boiling Point column. Go to File -> Save -> Method.

Reprocessing of results back to an instrument adjustment may be required following adjustment of the method. See Section 7.7.

7.5 Maintenance and Troubleshooting

Semiannual Preventive Maintenance

Replacing the Nafion dryer, replacing preconcentrator trap (s), servicing zero air generator, calibrating standard dilution measurement equipment

Changing a Column

See video

https://drive.google.com/file/d/1qvjzCigt_ZHIEnf9WBWZYHb7AlzZm70U/view?usp=sharing

Changing a Trap

The Unity and kori traps are made of glass and are very susceptible to breaking when being changed. Caution and proper handling are required. The trap pneumatic assembly will be loosened with a T20 Torques screwdriver. The metal trap connector will be loosened next while holding the trap pneumatic assembly in place. Once the metal connector is loosened the assembly can be pulled away from the trap. Next take the trap out by pulling it out slowly while twisting it with your two fingers. Keep your two fingers as close to the insertion hole as possible. Once loose then slowly pull out the trap a little at a time while keeping moving your fingers up close to the hole. When the trap is removed inspect O-rings inside the trap housing to see if they need replacement. Reverse the above steps for installing the new one and make sure that the trap gets put in all the way and that it seats into the O-ring. You can feel when this happens.

The new trap needs to be baked out and conditioned. This can be done by finding the "trap heat" method in the MIC software. This will include staged intervals of heating while

flowing carrier gas to eliminate contaminants. Stage 1 will be at 100 C for 10 minutes. Stage 2 will be 200 C for 10 minutes. Stage 3 will be at 320 C for 30 minutes.

Transfer Line Maintenance

The transfer line is a foam tube with a fused silica insert that connects the outlet from the TD trap to the front inlet of the GC. A column is fed from the GC through the fused silica insert to the outlet for the TD where it is tightened onto a union with a ferrule and a Swagelok nut. Occasionally, the transfer line will need maintenance.

In the Utah DAQ AutoGC program, there are two options for columns to connect the GC and TD. One is to connect an uncoated column via a union with the front end of the BP column. The ends of both columns will have ferrules that connect via the union. The uncoated column is fed through the fused silica insert of the transfer line into the TD. The second option is to feed the end of the BP column directly into the fused silica transfer line into the GC. The difference is that with the BP column fed directly, the compounds start separating within the transfer line, and it eliminates the need for the additional union and two ferrules inside the GC.

Changing Hydrogen Generator Gel

The Peak H2 generator has a gel life indicator on the screen and the gel should be changed once the gel life goes below 20%. The first step is to push the stop button on the main screen. Next push the menu button and then hit the next button once. The change gel option will appear and that will be pushed to depressurize the instrument. Remove the front panel of the instrument and inside is the gel cylinder on the bottom right of the instrument. Disconnect the tubing with the quick release and unscrew the cylinder. Once unscrewed, pull out the cylinder and unscrew the back end of the cylinder while holding it upright. Once unscrewed, pour the spent silica into a jar so it can be transported to the regeneration oven. Put new silica in the cylinder and reverse the steps. Make sure that the end of the cylinder lines up with the hole in the back of the instrument and that it seats into the O-ring. Once everything is hooked up, hit the confirm option on the screen. The instrument will then start to pressurize. It will take about 15-20 minutes to complete and then hit the start button to get Hydrogen back into the GC system. Sometimes it will not pressurize in time and an alarm will start. In this case the instrument is to be power cycled and it will complete a startup cycle. If it doesn't pressurize and the alarm sounds off again there is most likely a leak and diagnostic measures should be taken.

Troubleshooting

- Excess of humidity the sample stream: the additional water may interfere with chromatographic separation resulting in shortened retention times for target analytes on the light hydrocarbon channel (PLOT column analytes). Such retention time shifting may be transitory and retention times will usually stabilize and revert to their means once the ambient relative humidity levels decrease. The excess of humidity can also be reflected in the baseline of the FID detector which will show a dip every few minutes

Note: check for the presence of water droplets in the Kori and Unity windows, if there are some, open the chamber and wipe them with a lint-free wipe. Check the Kori purge flows and adjust them to 100 L/min if needed by rotating the dial located to the left (see Figure 5). Counterclockwise rotation increases the flow while the clockwise decreases the flow

Figure 5 Kori water condenser system

- **Retention Time Shifting**: RT shifts can occur due to a number of factors which include, but are not limited to: excess of humidity caused by insufficient purge flow, failure in the zero-air generator to deliver adequate dry gas, rain events causing atypical moisture in the ambient sample. Generally, causes retention time shifting on the PLOT column. Retention time shifting on the BP column will more likely be related to carry gas flow fluctuations. Retention time shifting can be a signal of leaks in the system
- **Excessive FID Noise or No peaks**
- **System Leaks**
- **Trap malfunction (Kori or Unity TD Traps):** Sorbents in the traps can degrade over time due to the hundreds of heating and cooling cycles experienced and eventually require replacement. Decreases in the GC-FID carbon response factors for the two calibrant gasses (propane and/or benzene or toluene) may indicate sorbents have degraded and the trap(s) is due for replacement. In addition, decreases in the recoveries of light (C2-C3) or heavy (C9- C10) hydrocarbons without decreases in C4-C7 hydrocarbons may also signal degradation of sorbents. Other signs of trap failure include sudden retention time shifts and poor chromatography (such as: broad, split, tailing peaks, unresolved unknown peaks)
- **Markes Heated Zone Failure**

7.6 Quality Control Checks, Logbook and Parameter Check List

Calibration Verification Standard (CVS)

The CVS is a daily analysis of a 1 ppmV standard comprised of 14 components blended at the site with humidified zero air. A list of compounds present in the CVS are shown in Table 1.

Under normal system operations, the CVS is scheduled to run daily and is followed by a method blank (see section 7.6.2). Instrument drift is considered acceptable when the percent recoveries for the compounds selected as a calibrant (propane or butane in the PLOT column and toluene in the BP column) compared to canister specific concentrations fall within 70-130 percent.

Note: when the CVS recoveries are not within the desired range, schedule a CVS rerun followed by a blank.

A precision check consisting of two back-to-back CVS samples is performed once a week to determine analytical reproducibility. A precision check is considered acceptable when the relative percent difference (RPD) between the two CVS runs is less than 25% for propane and benzene or Toluene, depending which is the calibrant for the BP column

Note: when a Precision Check does not meet the listed Acceptance Criteria, schedule another precision check and if it fails further investigation is necessary

Method Blank

A method blank is a humidified air matrix, void of contaminants, processed under the same conditions as samples. Results are assessed to determine if there is carry-over and contamination residing in the system. The daily blank is performed immediately after the daily CVS and is necessary to check for system cleanliness. Other blanks will run after multipoint calibrations and other system maintenance. The blank is considered acceptable when individual concentrations of target compounds are less than MDL or 0.5 ppbC if the MDLs are not estimated and Total Non-Methane Hydrocarbons (TNMHC) and Total Non-Methane Target Compounds (TNMTC) are less than 10.0 ppbC.

Note: when a blank check does not meet the listed Acceptance Criteria, schedule another blank check and if it fails further investigation is necessary.

Laboratory Calibration Standard (LCS)

The LCS is a 5 ppbV second source standard control sample that is scheduled every week to assess calibration accuracy. The LCS, like the CVS, is comprised of 14 components as shown in Table 1. The LCS sample is introduced directly into the instrument without blending with humidified zero air. The LCS can be used as an alternative to the CVS for calibration verification purposes if the CVS is faulty due to problems not related to the instrument, such as a bad CVS canister, or for individual compounds if the LCS can prove that the CVS is not a reliable measure of percent recovery (such as in the case of acetylene which can tend to have a lower recovery in the CVS). Since the LCS sample is injected directly into the instrument without passing through the blender, the resulting LCS recoveries can indicate if the calibration system and AutoGC are both drifting. Instrument drift is considered acceptable when the percent recoveries for propane and the

compound selected as a calibrant in the BP column (benzene or toluene) (compared to the initial calibration) fall within 70-130 percent.

Note: when the LCS check does not meet the listed Acceptance Criteria, schedule another LCS check and if it fails further investigation is necessary.

Retention Time Standard (RTS)

An RTS sample is scheduled every week and after any maintenance period to verify compound identification. This 57-component standard comprises all of the PAMS Program target compounds. It is analyzed to help assess elution order and retention time shifts in the chromatographic method.

Note: when the RTS check does not meet the listed Acceptance Criteria, schedule another RTS check and if it fails further investigation is necessary

Multipoint Calibration

A multipoint calibration is run to assess instrument linearity and establish response factors (RFs) used for quantitation. Multipoint calibrations are run following site start up, preventative maintenance, major repair or to correct the calibration of the instrument over time. At minimum, a multipoint calibration should be run once every six months. A four-point calibration curve is determined using mixtures of propane or butane and benzene or toluene at concentrations of approximately 1, 2, 10, and 50 ppbV. The response factor is then calculated for each level for both propane and benzene or toluene (depending which one is the calibrant), then averaged together to provide the overall response factors for each species. The calculated response factor for propane is applied to all compounds eluting on the PLOT column, and the calculated response factor for benzene or toluene is applied to all compounds eluting on the BP column. The multipoint calibration is considered acceptable by the operator when all four concentration levels are detected and the percent relative standard deviation (%RSD) of the four concentrations is less than 10%. The data validator considers the instrument to be stable if the final response factors are within 10% of the response factor calculated from the previous calibration.

Logbook and Checklist

Data validators review the operator logbook on a weekly basis at minimum or more frequently when quality control data or the site operator indicates there is an issue. The operator logbook and checklist are electronic logs kept by the site operator. The data validator is granted access to view the operator logbook and checklist via TeamViewer. Operator logbook entries typically summarize checks done on site or remotely, regular and preventive maintenance, problems, actions, repairs, and other items concerning instrument performance and data gathering. The operator checklist is a list of required parameters that the operator must check weekly. Data in the checklist can reveal a variety of instrument malfunctions.

7.7 Applying Chromatogram Corrections to Multiple Hours and Reloading Corrected Files

Reprocessing/optimizing

Sometimes there is an ongoing method issue affecting many hours which would require the same corrections across all affected hours. This is called a batch reprocess. A typical example is having the method shift on the PLOT column due to flow or temperature issues that cause the misidentification of several compounds.

Note: There is a very good tutorial video located: U:\PLAN\AMC\GC-Agilent+Markes\Training\EZChrom Overlays 480p (3)

- Use ORSAT MAX to select an hour that contains as many of the species as possible for the current day. So, if doing this early in the morning you may be limited. Look for an hour that either has the most in red values, thus being the maximum concentration so far, or an hour that has the most species available (for example, hours I or J, see Figure 6)

  Note: Run operators can only reprocess the current day's data

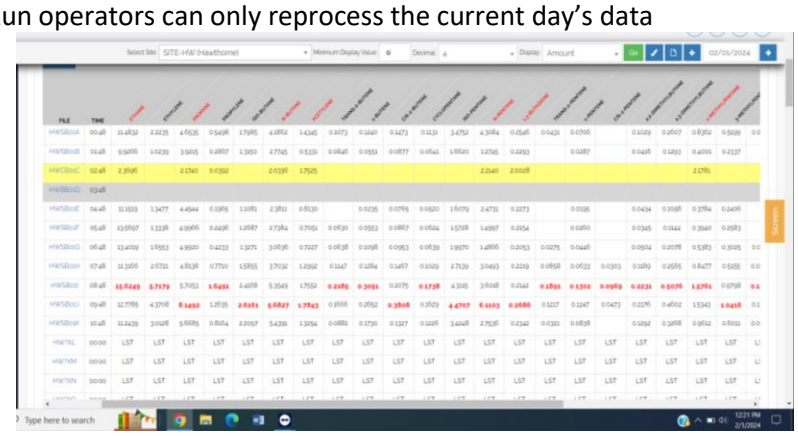


Figure 6. Example of a selected hour (red font) for reprocessing

- Go to the offline on the GC computer (remote or at the station)

NOTE: If a method is open on the offline, close it and then re-open the method (there should only be one method)

- Open the ambient chromatogram file for the selected hour File->open->data->current->selected hour (see Figure 7)

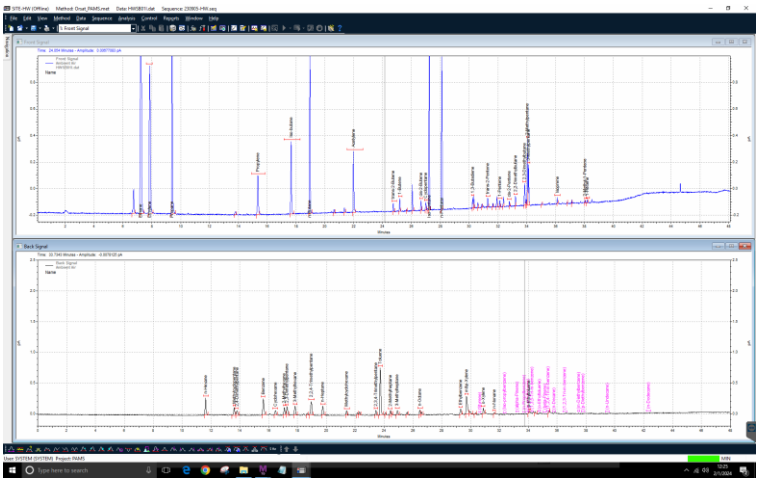


Figure 7. screen of the selected ambient hour

- Open the current folder where the .TXT files are loading ((Data (D:) or Datator (D:)) select and erase all .TXT files (Figure 8)

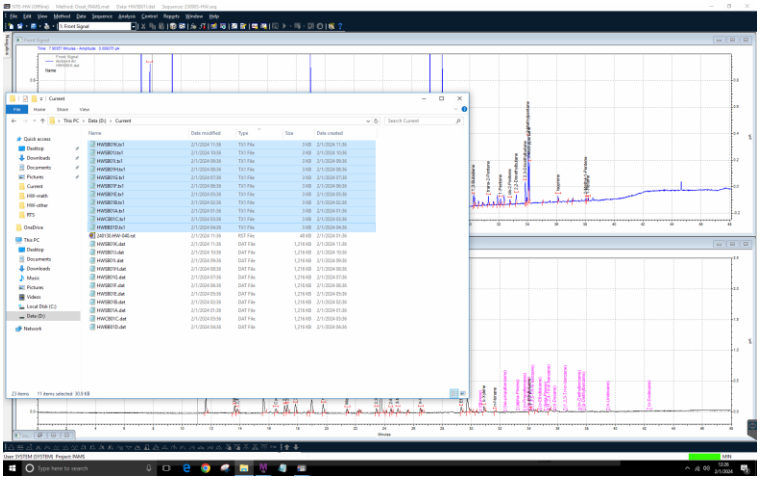


Figure 8. Erasing the TXT files in the current folder

- Right click on the front column "chromatogram" and click "add overlay". Look for the most recent CVS and RTS then overlay both files. Make sure that the overlapped files are normalized to the Trace 1 and add a Y offset that allows visualize the chromatograms. Include Name and RT window then click "apply all"
- Move both the CVS and RTS to align the peaks with the ambient file. Start aligning the peaks that are less affected by the drift such as n-pentane and make sure that all three traces match up with Three Amigos (2-3 Dimethylbutane, 2-Mehtylpentane, 3-Methylpentane) in the PLOT column (see Figure 9)

Note: the RT for the compounds in the BP column tend to drift less except for the problematic species that co-elute such as m and p-ethyltoluene and m-and p-diethylbenzene

Note: Using an ambient-spike run will help to verify some peaks that co-elute and are misidentified when their concentrations are low in the ambient air or exhibit poor peak shape (1,3-butadiene, styrene, cyclopentane, cis-2-butane

Figure 9. Overlapped chromatograms

- Adjust the Retention Time (RT) windows in the ambient trace and center them on their respective peaks

Note: By first clicking on the ambient trace, the icons on the bottom menu bar will appear. Select the Adjust Retention Time Window (Figure 10)

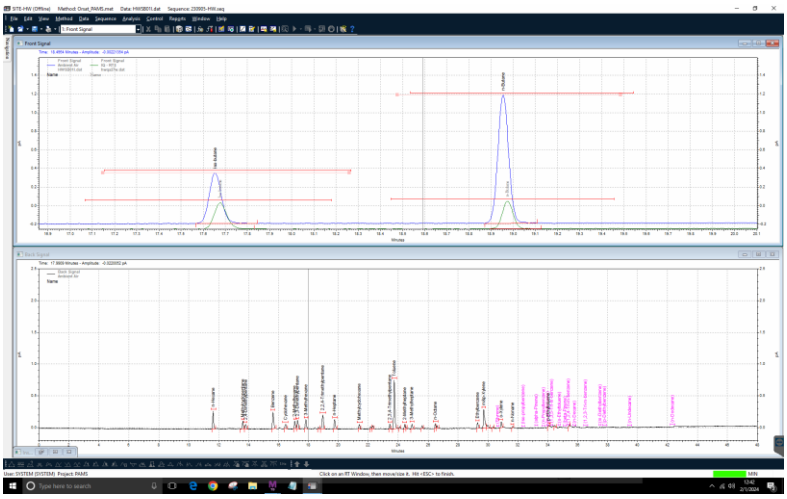


Figure 10. Retention Time window adjustment

- After adjusting every window, hit 'ESC'. Then when the pop-up window appears, select "Update RT"

- Repeat for BP column if a reprocessing for the BP column is required
    - Make sure the drop-down menu that says "back profile" (Figure 11)

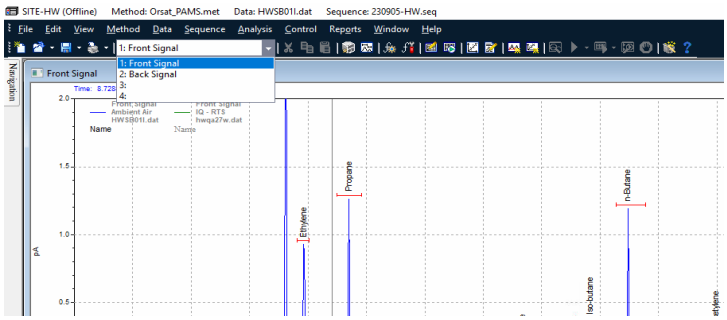


Figure 11. selecting the BP column if a reprocessing is required

- On the upper menu bar select "analyze".
- Add a comment like "reprocess" or "opt" then > apply that to all > then select 'OK' (Figure 12)

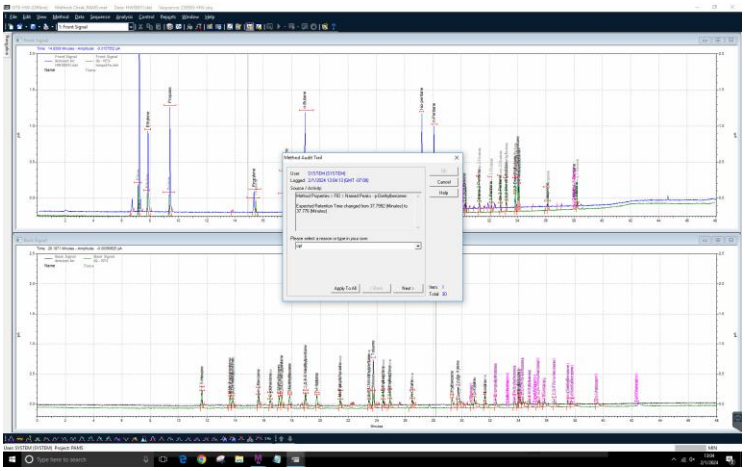


Figure 12. optimization screen

- On the same upper menu bar select Instrument wizard -> "create sequence"
- Select 'from existing data files' -> 'ok'
- Upload the current data files and Save as "RPOdateA-the hour reprocessed

- On the same upper menu bar select 'process sequence'. Name the result the same as the last steps name. Select 'all' for range > Select 'Ok'
- A bunch of windows will pop up and disappear, once the system is done, the files are successfully reprocessed
- After the sequence has been optimized go to Orsat max (on the GC computer) and upload the new .TXT files
- Record what you did in the station log

Reloading Corrected Files

- After completing all corrections and batch reprocesses, ensure all corrected hours have a new modified date for *.dat and *.tx1 files
- Navigate to the ORSAT MAX site page
- On the left toolbar of the ORSAT MAX page, click on the Data Manager Icon -> Upload EZChrom
- Click "Select Data Files"
- Navigate to the relevant AutoGC Data folder and select the corrected *.tx1 files. Click "Open" (Figure 13)

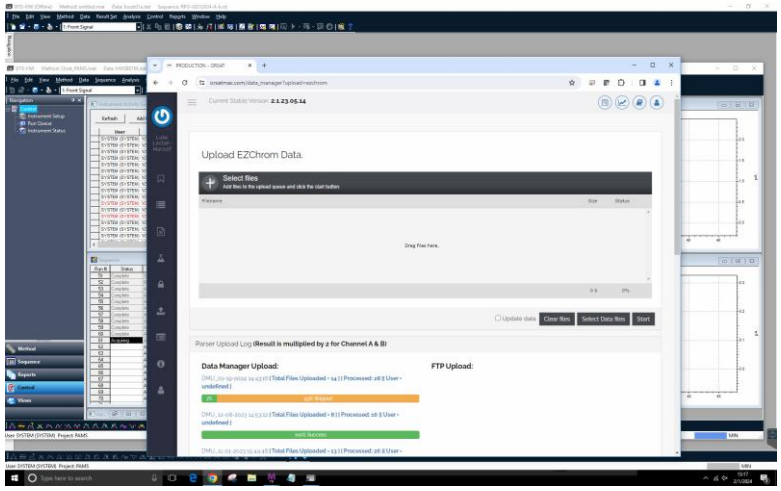


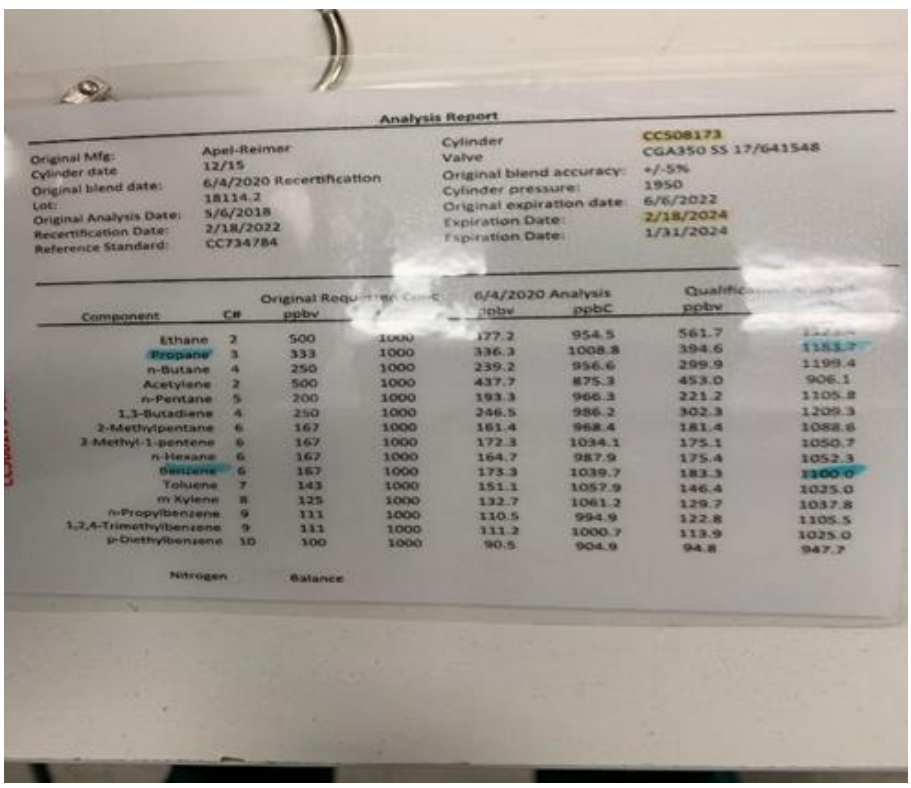
Figure 13. screen to select the files that will be reloaded

- **Select the checkbox next to "Update Data". Failure to do this will result in the data from the corrected *.tx1 files not being properly uploaded**
- Click "Start"

- Once the upload process is complete, navigate to a amount or retention time quicklook page for a corrected hour (depending on what correction was performed) and verify that the data matches the corrected *.tx1 or *.dat files
- Note: close all windows in the Offline screen so that those files can get zipped with that data at the end of the day

7.8 Replacing the Canisters

- Take a new canister that will need to be replaced (CVS, LCS, or RTS) from the Tech Center to the site
- Close old canister tight and undo the fitting
- Install a new canister. You do not have to wrench them down very hard. Finger snug and then another quarter to half turn. Absolutely NO SNOOP (Make it a point to plan a return to site visit a day or two later to see if there is a significant change indicating a leak
- Take photos of the paperwork included with the new canister (CVS, LCS and RTS) (Figure 14 and Figure 15) and Save them in the corresponding site folder U:\PLAN\AMC\GC-Agilent+Markes\Data\MDVR\Cylinder documentation
  Note: The Photos should be named (cylinder type.month.day.year.image #) CVS.1.1.24.image1



Analysis Report

Original Mfg: Apel-Reimer
Cylinder date 12/15
Original blend date: 6/4/2020 Recertification
Lot: 18114.2
Original Analysis Date: 5/6/2018
Recertification Date: 2/18/2022
Reference Standard: CC734784

Cylinder CC508173
Valve CGA350 SS 17/641548
Original blend accuracy: +/-5%
Cylinder pressure: 1950
Original expiration date: 6/6/2022
Expiration Date: 2/18/2024
Expiration Date: 1/31/2024

Component	C#	Original Requ		6/4/2020 Analysis ppbv	ppbC	Qualific ppbv	
Ethane	2	500	1000	177.2	954.5	561.7	
Propane	3	333	1000	336.3	1008.8	394.6	1181.7
n-Butane	4	250	1000	239.2	956.6	299.9	1199.4
Acetylene	2	500	1000	437.7	875.3	453.0	906.1
n-Pentane	5	200	1000	193.3	966.3	221.2	1105.8
1,3-Butadiene	4	250	1000	246.5	986.2	302.3	1209.3
2-Methylpentane	6	167	1000	161.4	968.4	181.4	1088.6
2-Methyl-1-pentene	6	167	1000	172.3	1034.1	175.1	1050.7
n-Hexane	6	167	1000	164.7	987.9	175.4	1052.3
Benzene	6	167	1000	173.3	1039.7	183.3	1100.0
Toluene	7	143	1000	151.1	1057.9	146.4	1025.0
m Xylene	8	125	1000	132.7	1061.2	129.7	1037.8
n-Propylbenzene	9	111	1000	110.5	994.9	122.8	1105.5
1,2,4-Trimethylbenzene	9	111	1000	111.2	1000.7	113.9	1025.0
p-Diethylbenzene	10	100	1000	90.5	904.9	94.8	947.7

Nitrogen Balance

Figure 14. New canister documentation

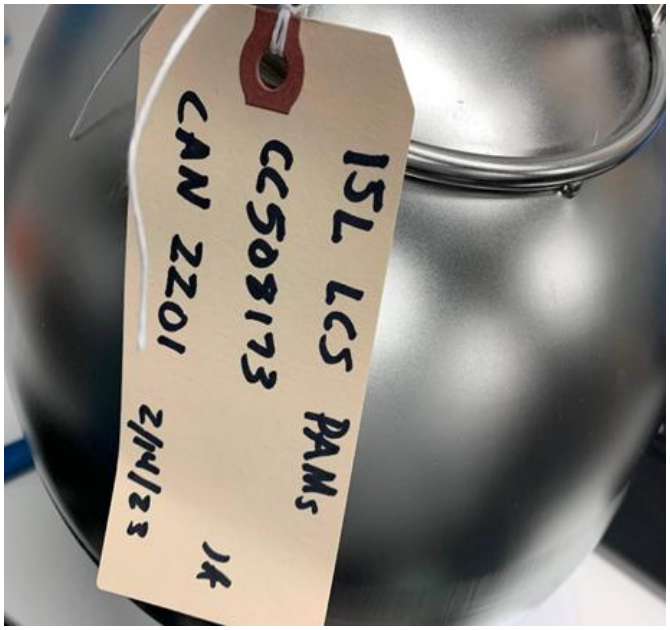


Figure 15. New canister Information

- On the corresponding GC computer, open the checklist excel sheet, update highlight (yellow) the new concentration the proper cylinder information in left column (Figure 16)

24	H2 cylinder outlet, psi	90	90	91	91	91
25						
26	Gas-01 Carrier	60+	60+	60+	60+	60+
27	Gas-01 Purge	60	60	60	59	59
28	Gas-02 Carrier	25	25	25	25	25
29	Gas-02 Purge	59	59	59	59	57
30						
31	15L LCS CC177206-0125-3186	22	22	20	20	17
32	15L CVS CC524271-0325-3180	9	32	30	29	28
33	RTS cylinder/canister, psi	5	30	30	30	29
34	**RTS CYLINDER VALVE CLOSED**					
35	TOC Catalyst	warm	warm	warm	warm	warm
36	Zero Air Catalyst	warm	warm	warm	warm	warm
37	Magnahelix-	0.4	0.4	0.4	0.39	0.39
38	Sample lines	warm	warm	warm	warm	warm
39	He Purifier	warm	warm	warm	warm	warm

Figure 16. Check List excel sheet

- Send the new canister information to the person in charge to input the data in Max (currently, Cristina)

7.9 Manifold Residence Time

To estimate the manifold residence time, use the following google sheet template:

https://docs.google.com/spreadsheets/d/1jgtOIU1ifSynYTKaUEY_9tOK6Osqi9R2/edit?usp=drive_link&ouid=112066211249095730412&rtpof=true&sd=true

8.0 REFERENCES

Photochemical Assessment Monitoring Station (PAMS) Network Quality Assurance Project Plan (QAPP) for Utah Air Monitoring

Texas Commission on Environmental Quality, Monitoring Division SOP DV-002-VOC, Revision 0, Automated Gas Chromatograph (AutoGC) Data Collection, Management, Validation, and Reporting, July 2018

Automated Gas Chromatograph (AUTOGC) Data Management and Validation in Orsat's Merlin AutoGCXplorer (MAX) Database SOP by Andrea Connelly, March 2021

U.S.EPA. Technical Assistance Document for Sampling and Analysis of Ozone Precursors for the photochemical Assessment Monitoring Stations Program, Revision 3. April 2023. Available at:

https://www.epa.gov/system/files/documents/2023-05/TAD%20R3%20May%202023.pdf

U.S EPA. Quality Assurance Project Plan for the Photochemical Assessment Monitoring Stations (PAMS) Required Site Network for Speciated Volatile, Organic Compounds, Carbonyls,a nd Meteorological Parameters Including Mixing Layer Height, Revision 1. April 2023

https://www.epa.gov/system/files/documents/2024-04/pams-qapp-r1-may-2023.pdf

9.0 REVISION HISTORY

Revision 1, Effective Date: April 01, 2023

Revision 2. Effective Date: May 01, 2024

APPENDIX A. UTAH DAQ MASTER AUTOGC COMPOUND LIST

Priority Compounds		Optional Compounds	
1,2,3-trimethylbenzene [a]	m-ethyltoluene [a]	1,3,5-trimethylbenzene	isopropylbenzene [b]
1,2,4-trimethylbenzene [a]	n-butane	1-pentene	m-diethlybenzene
1-butene	n-hexane [b]	2,2-dimethylbutane	methylcyclohexane
2,2,4-trimethylpentane [b]	n-pentane	2,3,4-trimethylpentane	methylcyclopentane
benzene [a,b]	o-ethyltoluene [a]	2,3-dimethylbutane	n-decane
cis-2-butene	o-xylene [a,b]	2,3-dimethylpentane	n-heptane
ethane [c]	p-ethyltoluene	2,4-dimethylpentane	n-nonane
ethylbenzene [a,b]	propane	2-methylheptane	n-octane
ethylene	propylene	2-methylhexane	n-propylbenzene [a]
isobutane	styrene [a,b]	2-methylpentane	n-undecane
isopentane	toluene [a,b]	3-methylheptane	p-diethylbenzene
isoprene	trans-2-butene	3-methylhexane	trans-2-pentene
m&p-xylenes [a,b]	total non-methane organic carbon (TNMOC)	3-methylpentane	**α/β-pinene**
		acetylene	1,3 butadiene [b]
		cis-2-pentene	cyclohexane
		cyclopentane	

Source: Revisions to the Photochemical Assessment Monitoring Stations Compound Target List. U.S. EPA, November 20, 2013

[a] Important SOAP (Secondary Organic Aerosols Precursor) Compounds

[b] HAP (Hazardous Air Pollutant) Compounds. [c] Non-reactive compounds, not considered to be VOC for regulatory purposes